

Current Position

Postdoctoral Research Fellow Carleton University Supervisor: Millard K 2026–

Work Experience

Research Associate Carleton University Supervisor: Millard K 2025–2026

Education

Ph.D	University of Toronto	2020–2026
	Supervisors: Fortin M-J, James PMA	
M.Sc.	Concordia University	2017–2019
	Supervisor: Brown GE	
B.Sc (Hons)	University of Toronto	2013–2017
	Supervisor: Malcolm JR	

Publications

In Preparation

13. Goldman JA*, Millard K, Pelletier N, Thompson DK, Pontone N, Guindon L. Severe droughts are putting Canadian peatland carbon stocks at risk: an analysis of fire selectivity and behavior in peatlands over the extreme 2023 fire season.
12. Wilkinson SL, Leverkus S, Goldman JA*, Butchart C, Flannigan M. Multi-year overwintering fires in Boreal Canada.
11. Wilkinson SL, Goldman JA*, Krieger-Pottruff E, James PMA. Legacies of mountain pine beetle defoliation influence burn severity in western Canada
10. Goldman JA*, Fortin MJ, James PMA. The impact of spruce budworm defoliation legacies on wildfire burn severity and post-fire recovery.

In Review

9. Goldman JA*, Fortin MJ, James PMA. Within and among year moisture availability influences wildfire burn severity in the boreal forest. *Fire Ecology*.

Published

8. Goldman JA*, Fortin MJ, James PMA. 2025. Snow dynamics and forest structure interact to increase wildfire burn severity in the boreal forest. *Forest Ecology and Management*.
7. Messaoud Y, Reid A, Tchebakova NM, Goldman JA*, Hofgaard A. 2022. The historical complexity of tree height growth dynamic associated with climate change in western north america. *Forests*, 13(5): 738.
6. Crane AL, Bairos-Novak KR, Goldman JA*, Brown GE. 2022. Chemical disturbance cues in aquatic systems: a review and prospectus. *Ecological Monographs*, 92(1): e01487.
5. Goldman JA*, Crane AL, Feyten LEA, Collins E, Brown GE. 2022. Disturbance cue communication is shaped by emitter diet and receiver background risk in Trinidadian guppies. *Current Zoology*, 68(4): 433440.
4. Brown GE, Demers EM, Goldman JA*, Singh A, Chivers DP, Ferrari MCO. 2020. Unpredictable risk enhances induced neophobia in northern red-bellied dace. *Animal Behaviour*, 168: 121127.
3. Goldman JA*, Desormeaux IS, Brown GE. 2020. Disturbance cues function as sources of risk assessment information under natural conditions. *Freshwater Biology*, 65(5): 981986.
2. Goldman JA*, Feyten LEA, Ramnarine IW, Brown GE. 2020. Sender and receiver experience with risk alters the response to disturbance cues by Trinidadian guppies. *Current Zoology*, 66(3): 255261.
1. Goldman JA*, Singh A, Demers EM, Feyten LEA, Brown GE. (2019) Does donor group size matter? The response of Trinidadian guppies and Convict cichlids to disturbance cues from conspecific and heterospecific donors. *Canadian Journal of Zoology*, 97(4) 319-325. *Selected as Editors Choice

Presentations

- 2026 The impact of spruce budworm defoliation legacies on wildfire burn severity and post-fire recovery. Wildfire Canada
- 2024 Comparing forest recovery trajectories following wildfire in the presence and absence of spruce budworm defoliation using remotely sensed data. IALE-North America Annual Meeting, Oklahoma City, Oklahoma.
- 2023 The effects of snowmelt timing on wildfire burn severity in the boreal forest. IALE-North America Annual Meeting, Riverside, California.
- 2022 Time-lagged and cumulative effects of moisture availability on burn severity in the boreal forest. Wildland Fire Canada Conference, Edmonton, AB.
- 2022 Trends in burn severity across an east-west gradient. Ecological Society of America Annual Meeting, Montreal, QC.
- 2022 Spatial and temporal trends in landscape patterns of burn severity in the boreal shield ecozone of Ontario. North American Forest Ecology Workshop (virtual).
- 2019 Quality over quantity: Sender diet quality, but not quantity, shapes the production of disturbance cues. Canadian Society for Ecology and Evolution Fredericton, NB
- 2019 Predation risk experienced by senders and receivers alters the response of Trinidadian guppies to disturbance cues. Canadian Conference for Fisheries Research London, ON
- 2018 Evidence for species-specificity in disturbance cues. Ecology and Evolutionary Ethology of Fishes (EEEEF) Montreal, QC *Awarded Best Poster
- 2017 Concordia Scientific Communication Poster Symposium Montreal, QC
- 2017 University of Toronto Research Opportunity Program Poster Symposium Toronto, ON

Scholarships & Awards

- 2024-2025 School of Graduate Studies Doctoral Completion Award
- 2023-2024 Queen Elizabeth II Graduate Scholarship in Science and Technology
- 2022 North American Forest Ecology Workshop Student Presentation Award 3rd Place
- 2020-2021 Edward H. Buckley and Grace B. Buckley Graduate Scholarship in Forestry
- 2020-2021 GSEF Faculty of Forestry Award
- 2020-2021 Louis J. Nozzollilo Fellowship in Forestry Research
- 2020-2023 NSERC Postgraduate Scholarship Doctoral
- 2019-20 Queen Elizabeth II Graduate Scholarship in Science and Technology
- 2019 Concordia SGS Conference and Exhibition Travel Award
- 2019 Clemens-Rigler Travel Award, CCFFR
- 2018 Gerry Fitzgerald Award, Best Poster Presentation at EEEF 2018
- 2018-19 Concordia Graduate Scholarship Natural Sciences and Engineering Research
- 2017 Edward H. Buckley and Grace B. Buckley Undergraduate Scholarship in Forestry
- 2017 NSERC Undergraduate Summer Research Award

Teaching

2023	FOR3012: Analytical Methods in Forestry Statistics (graduate) Teaching Assistant	University of Toronto
2020	BIO120: Adaptation and Biodiversity Teaching Assistant	University of Toronto
2019	BIOL 4259/9915: Research Hypothesis Testing (undergrad/grad) Teaching Assistant	Western University
2018	BIOL 227: Laboratory Studies of Organisms Teaching Assistant	Concordia University
2017	BIOL 201: Introductory Biology Teaching Assistant	Concordia University

Lectures & Workshops

2025	Harnessing The Power of Google Earth Engine with Python University of Toronto
2023	Multivariate Analysis and Ordination University of Toronto

Professional Development

2023	Short Course on Forecasting for Decision-Making: An Epidemiological & Ecological Perspective (Invited) Canadian Ecological Forecasting Initiative & Fields Institute, University of Toronto
2023	Structural Equation Modelling Summer School Université de Sherbrooke

Skills & Tools

Programming:	R, Python, JavaScript, bash
Project Management:	git, uv, uvr, Quarto
Data Management:	duckDB, PostgreSQL
GIS:	Google Earth Engine, QGIS, ArcPy